PROBABILISTIC ARTIFICIAL INTELLIGENCE

Homework 1

Probability & Bayesian linear regression

Exercises 1–4

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Exercise 0	Exercise 1	Exercise 2	Exercise 3	Exercise 4
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6	10	21	29	37

Exercise 1(a)

Marginals of a Gaussian vector

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Statement: The marginal distribution of any subset of components of X is not Gaussian.

Working

$k \leq n$ $\quad I \subset \{1, \ldots, n\}$ $\quad I = \{i_1, \ldots, i_k\}$.

$A X = X_I = (X_{i_1}, \ldots, X_{i_k})$ i-th pos $\quad A X = X_I$

$e_i = (0, \ldots, \underset{i}{1}, 0, \ldots 0)$

$e_i^T X = X_i$ $\quad A = \begin{bmatrix} e_{i_1}^T \\ \vdots \\ e_{i_k}^T \end{bmatrix}$

Exercise 1(a)

Linear combinations of jointly Gaussian variables

Statement: Linear combinations of jointly Gaussian random variables are Gaussian.

$$X = (X_1, \ldots, X_n)^\top, Y = \sum_{i=1}^{n} a_i X_i$$

$X \sim N(\mu, \Sigma)$

$AX \sim N(A\mu, A\Sigma A^\top)$

Working

$AX = Y$

$a = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$

$a^\top X = \sum_i a_i X_i = Y$

Exercise 1(b)

A nonlinear expression of Gaussian variables

Let X, Y and Z be independent standard normal random variables.

$$T = \frac{X + e^Z Y}{\sqrt{1 + e^{2Z}}}$$

Statement: T is not Gaussian.

False

$X, Y, Z \overset{iid}{\sim} N(0,1)$.

Ex 0: a).

T, Z

$p(T \mid Z = z) = g(T, \cdot)$

$p(T) = \int p(T \mid Z = z)$

Working

$$p(T \mid Z = z) = p\left(\frac{X + e^z Y}{\sqrt{1 + e^{2z}}} \,\middle|\, Z = z \right)$$

$$= p\left(\frac{X + e^z Y}{\sqrt{1 + e^{2z}}} \right)$$

$B := X + e^z Y \sim N(0, 1 + e^{2z})$

$$T \mid Z = z = \frac{B}{\sqrt{1 + e^{2z}}} \sim N(0,1)$$

$T \sim N(0,1)$

Exercise 1(b)

A random angle

Let X and Y be independent standard normal variables. Let α be independent of them, with

$$P(\alpha=0)=P\left(\alpha=\frac{\pi}{2}\right)=\frac{1}{2}.$$

$$Z:=X\cos\alpha+Y\sin\alpha\sim\mathcal{N}(0,1)$$

True

Working

When are the sum and difference independent?

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} 1 & \alpha \\ \alpha & 1 \end{bmatrix}\right), |\alpha| \leq 1$$

$1 - \alpha^2 < 0$

$|\alpha| > 1.$

Statement: X + Y and X − Y are independent if and only if α = 0.

Working

$$A \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} U \\ V \end{bmatrix} \qquad A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} U \\ V \end{bmatrix} \sim N\left(A0, A \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix} A^T\right)$$

$$\begin{bmatrix} 2+2\alpha & 0 \\ 0 & 2+2\alpha \end{bmatrix}$$

Exercise 1(c)

Parameters of independent binary variables

Let $X_1, \ldots, X_n$ be mutually independent binary random variables.

Statement: Their joint distribution requires $\Theta(2^n)$ parameters.

Working

$$p(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} p(X_i = x_i)$$

A simpler Gaussian coordinate system

Let $Z \sim N(\mu, \Sigma)$ be an n-dimensional Gaussian vector.

Statement: We can find an invertible A so that describing the distribution of AZ needs only O(n) parameters.

Working

μ, Σ $\quad \frac{n(n+1)}{2}$

$\Sigma = P D P^T$

$\Sigma' = A \Sigma A^T$ $\quad A := P^T$ $\quad = D$

$P^T Z \sim N(P^T \mu, P^T P D D^T P)$

Exercise 2(a)

Marginals of a multivariate Gaussian

Let $x \sim N(\mu, \Sigma)$, with Σ positive definite. Partition x into x_A and x_B.

$$p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2} (x-\mu)^\top \Sigma^{-1} (x-\mu) \right], x \in \mathbb{R}^d$$

$$\mu = \begin{bmatrix} \mu_A \\ \mu_B \end{bmatrix}, \Sigma = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}$$

Prove that every marginal is Gaussian. Identify the law of x_A.

Working

$$p(x_A) = \int_{x_B} p(x_A, x_B) \, dx_B$$

Joint density and block notation

Solution: Consider $x = \begin{bmatrix} x_A \\ x_B \end{bmatrix}$, $\mu = \begin{bmatrix} \mu_A \\ \mu_B \end{bmatrix}$, and $\Sigma = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}$, the joint distribution:

$$p(x) = p(x_A, x_B) = \frac{1}{Z} \exp\left(-\frac{1}{2} \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix}^T \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}^{-1} \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix} \right).$$

The following notations can ease the computation:

$$V = \begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix} = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}^{-1}, \quad \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix} = \begin{bmatrix} \Delta_A \\ \Delta_B \end{bmatrix}$$

Completing the square and integrating

$$p(x_A) = \frac{1}{Z}\int_{x_B} \exp\left(-\frac{1}{2}\begin{bmatrix}\Delta_A\\ \Delta_B\end{bmatrix}^T \begin{bmatrix}V_{AA} & V_{AB}\\ V_{BA} & V_{BB}\end{bmatrix}\begin{bmatrix}\Delta_A\\ \Delta_B\end{bmatrix}\right) dx_B \qquad \text{marginalise out } x_B$$

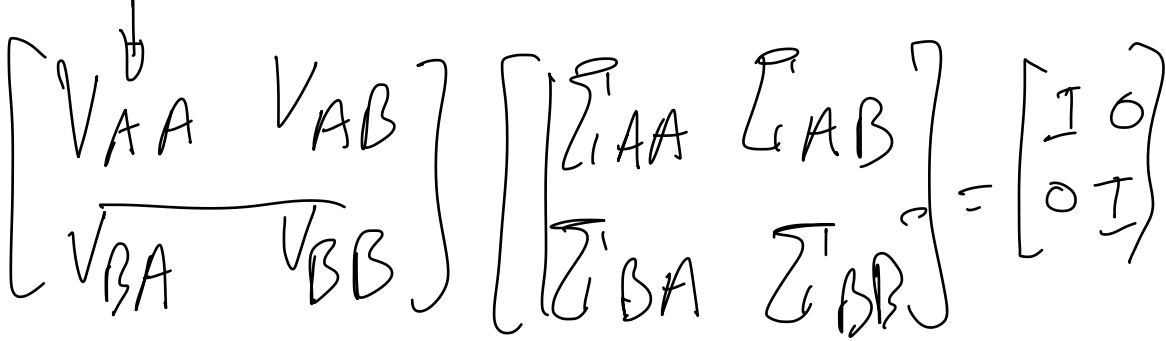
$$= \frac{1}{Z}\exp\left(-\frac{1}{2}\left[\Delta_A^T(V_{AA} - V_{AB}V_{BB}^{-1}V_{BA})\Delta_A\right]\right)\cdot \int_{x_B} \exp\left(-\frac{1}{2}\left[(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)^T V_{BB}(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)\right]\right) dx_B. \qquad \text{complete the square}$$

Note that the integrand is an unnormalized Gaussian pdf, so the value of the integral is $\sqrt{\det(2\pi V_{BB}^{-1})}$, which is a constant respect to x_A. We can therefore simplify it to

$$p(x_A) = \frac{1}{Z_A}\exp\left(-\frac{1}{2}\left[\Delta_A^T(V_{AA} - V_{AB}V_{BB}^{-1}V_{BA})\Delta_A\right]\right) = \frac{1}{Z_A}\exp\left(-\frac{1}{2}\left[\Delta_A^T\Sigma_{AA}^{-1}\Delta_A\right]\right) \quad \mathcal{N}(\mu_A, \Sigma_{AA})$$

$$\Sigma_{AA}^{-1} = V_{AA} - V_{AB}V_{BB}^{-1}V_{BA}$$

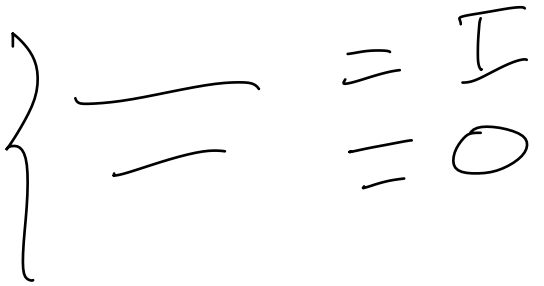
The covariance block identity



The last equality holds because:

$$\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix} = \begin{bmatrix} I & V_{AB}V_{BB}^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} V_{AA} - V_{AB}V_{BB}^{-1}V_{BA} & 0 \\ 0 & V_{BB} \end{bmatrix} \begin{bmatrix} I & 0 \\ V_{BB}^{-1}V_{BA} & I \end{bmatrix},$$

$$\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix}^{-1} = \begin{bmatrix} I & 0 \\ -V_{BB}^{-1}V_{BA} & I \end{bmatrix} \begin{bmatrix} (V_{AA} - V_{AB}V_{BB}^{-1}V_{BA})^{-1} & 0 \\ 0 & V_{BB}^{-1} \end{bmatrix} \begin{bmatrix} I & -V_{AB}V_{BB}^{-1} \\ 0 & I \end{bmatrix}.$$



The upper-left entry of $\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix}^{-1}$ is $(V_{AA} - V_{AB}V_{BB}^{-1}V_{BA})^{-1}$. Note that

$$\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix}^{-1} = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix},$$

hence

$$\Sigma_{AA} = (V_{AA} - V_{AB}V_{BB}^{-1}V_{BA})^{-1}.$$

Using similar arguments, we can also show that

$$V_{BB} = (\Sigma_{BB} - \Sigma_{BA}\Sigma_{AA}^{-1}\Sigma_{AB})^{-1}.$$

Exercise 2(b)

Conditionals of a multivariate Gaussian

Use the same partition of $x \sim N(\mu, \Sigma)$, with Σ positive definite.

Prove that conditioning on a subset of a jointly Gaussian vector gives a Gaussian distribution.

Find the mean and covariance of $x_B \mid x_A$.

Working

The conditional Gaussian kernel

Solution: Noting that $p(x_A)$ is a constant with respect to x_B, we have

$$p(x_B|x_A) = \frac{p(x_A, x_B)}{p(x_A)} =$$

$$= \frac{1}{Z'} \exp\left(-\frac{1}{2}\begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix}^T \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}^{-1} \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix}\right) \qquad p(x_A) \text{ absorbed into } Z'$$

$$= \frac{1}{Z'} \exp\left(-\frac{1}{2}\left[\Delta_A^T (V_{AA} - V_{AB}V_{BB}^{-1}V_{BA})\Delta_A\right]\right) \cdot$$

$$\cdot \exp\left(-\frac{1}{2}\left[(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)^T V_{BB}(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)\right]\right) \qquad \text{complete the square}$$

$$= \frac{1}{Z''} \exp\left(-\frac{1}{2}\left[(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)^T V_{BB}(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)\right]\right) \qquad \text{1st factor free of } x_B$$

Conditional mean and covariance

$x_B | x_A$

$$x_B|x_A \sim \mathcal{N}(\underbrace{\mu_B - V_{BB}^{-1}V_{BA}(x_A - \mu_A)}_{=\mu_{B|A}}; \underbrace{\Sigma_{BB} - \Sigma_{BA}\Sigma_{AA}^{-1}\Sigma_{AB}}_{=\Sigma_{B|A}=V_{BB}^{-1}})$$

$$\sim N(\mu_B + \Sigma_{BA}\Sigma_{AA}^{-1}(x_A - \mu_A), \Sigma_{BB} - \Sigma_{BA}\Sigma_{AA}^{-1}\Sigma_{AB})$$

$$\mu_{B|A} = \mu_B + \Sigma_{BA}\Sigma_{AA}^{-1}(x_A - \mu_A)$$

Characteristic functions

For a random vector, the characteristic function is defined by

$$\varphi_X(t) = \mathbb{E}[\exp(it^\top X)], X, t \in \mathbb{R}^d.$$

The characteristic function completely identifies a distribution.

For a multivariate Normal distribution:

$$X \sim \mathcal{N}_d(\mu, \Sigma) \Leftrightarrow \forall t \in \mathbb{R}^d, \varphi_X(t) = \exp\left(it^\top \mu - \frac{1}{2} t^\top \Sigma t\right).$$

$\varphi_{Ax+\mu}(t)$

Exercise 2(c)

An affine transformation

Let $X = (X_1, \ldots, X_d)$ be a d-dimensional standard Gaussian random vector, that is, $X \sim \mathcal{N}_d(0, I)$.

Define $Y = AX + \mu$, where A is a $d \times d$ matrix and $\mu \in \mathbb{R}^d$.

What is the distribution of Y?

$s := (t^T A)^T$

Working

$$\varphi_Y(t) = \mathbb{E}\left[\exp\{i t^T (AX + \mu)\}\right]$$
$$= \mathbb{E}\left[\exp\{i t^T A X\}\right] \exp\{i t^T \mu\}$$
$$= \varphi_X(s) \exp\{i t^T \mu\}.$$
$$= \exp\left\{-\frac{1}{2} s^T s\right\} \exp\{i t^T \mu\} = \exp\left\{-\frac{1}{2} t^T A A^T t + i t^T \mu\right\}.$$

$Y \sim N(\mu, AA^T)$

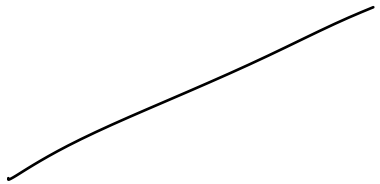
Exercise 2(d)

Another linear transformation

$$Y \sim \mathcal{N}\left(\mu, A A^{\top}\right), B \in \mathbb{R}^{r \times d}$$

What is the distribution of BY?

Working



Exercise 2(e)

Independence after a linear transformation

Let X and Y be independent $N(0, 1)$ random variables.

Statement: $X + Y$ and $X - Y$ are independent.

Working

$$A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix} \sim N\left(0, \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\right)$$

Exercise 2(f)

A conditional Gaussian with numerical inputs

$$X \sim \mathcal{N}\left(\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}5 & 1\\1 & 4\end{bmatrix}\right)$$

$$\begin{bmatrix}Y\\Z\end{bmatrix} \sim N(\mu', \Sigma')$$

$$Y = X_1 + X_2, Z = X_1 - X_2$$

Find the mean and variance of Y given Z = 0.

$\mu_{Y|Z=0}$, $\Sigma_{Y|Z}$

Working

$$A = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}$$

$$A\Sigma A^T = \begin{bmatrix}10 & 2\\2 & 7\end{bmatrix} = \Sigma' \qquad \mu' = \begin{bmatrix}2\\0\end{bmatrix}.$$

$$\mu_{Y|Z=0} = 2 - 1 \cdot \frac{2}{7}(0-0) = 2$$

$$\Sigma_{Y|Z} = 11 - 1 \cdot \frac{2}{7} \cdot 1 = \frac{77-2}{7} = \frac{76}{7}.$$

Exercise 2(f)

Conditional mean and variance

Find the mean and variance of Y conditional on Z = 0.

Working

Exercise 2(g), statement 1

Marginal covariance in Bayesian regression

$$y = Xw + \varepsilon, w \sim \mathcal{N}(\mu, \lambda^{-1} I_d)$$

$$\varepsilon \sim \mathcal{N}(0, \lambda^{-1} I_n), \varepsilon \text{ independent of } w$$

$X \in \mathbb{R}^{n \times d}$ is fixed and $\lambda > 0$ is known. Is the following statement true?

$$\Sigma_y = Var(y \mid X, \mu, \lambda) = \lambda^{-1} X X^T + I_n$$ FALSE

Working

$$Var[y \mid X, \mu, \lambda) = Var[Xw + \varepsilon \mid X, \mu, \lambda].$$

$$= X Var[w \mid X, \mu, \lambda] X^T + Var[\varepsilon \mid X, \mu, \lambda]$$

$$= \lambda^{-1} X X^T + \lambda^{-1} I = \lambda^{-1}(XX^T + I_n)$$

The marginal covariance of y

Solution: Let $f = Xw$. Since $w \mid \mu, \lambda \sim \mathcal{N}(\mu, \lambda^{-1} I_d)$, we see from lecture 1, slide 52 that

$$f \mid X, \mu, \lambda \sim \mathcal{N}(X\mu, \lambda^{-1} XX^\top). \qquad \mathrm{Cov}(Xw) = X(\lambda^{-1} I_d)X^\top$$

Thus,

$$y \mid X, \mu, \lambda \sim \mathcal{N}(X\mu, \lambda^{-1} XX^\top + \lambda^{-1} I_n). \qquad \epsilon \perp f,\ \epsilon \sim \mathcal{N}(0, \lambda^{-1} I_n)$$

Exercise 2(g), statement 2

Estimating the prior mean by maximum likelihood

Use the same model, with $\Sigma_y = \lambda^{-1}(XX^\top + I_n)$.

Is this the maximum likelihood estimate of the hyperparameter μ?

$$\hat{\mu} = \left(X^\top \Sigma_y^{-1} X \right)^{-1} X^\top \Sigma_y^{-1} y$$

Working

The negative log likelihood

Solution: We have

$$\hat{\mu}_1 = \arg\max_{\mu} p(y \mid X, \mu, \lambda) = \arg\max_{\mu} \log \mathcal{N}(X\mu, \Sigma_y). \qquad \log \text{ is monotone}$$

We can simplify

$$\begin{aligned} -\log \mathcal{N}(X\mu, \Sigma_y) &= \frac{1}{2}(y - X\mu)^\top \Sigma_y^{-1}(y - X\mu) + \text{const} \\ &= \frac{1}{2}\mu^\top X^\top \Sigma_y^{-1} X\mu - \mu^\top X^\top \Sigma_y^{-1} y + \text{const.} \qquad \text{drop terms free of } \mu \end{aligned}$$

The gradient and the estimator

(Note that $\mu^\top X^\top \Sigma_y^{-1} y$ and $y^\top \Sigma_y^{-1} X\mu$ are both scalars so :

$$\mu^\top X^\top \Sigma_y^{-1} y = (\mu^\top X^\top \Sigma_y^{-1} y)^\top = y^\top \Sigma_y^{-1} X\mu$$

and Σ_y^{-1} is a symmetric matrix).

Taking the gradient with respect to μ, we obtain

$$\nabla_\mu \left(\frac{1}{2} \mu^\top X^\top \Sigma_y^{-1} X\mu - y^\top \Sigma_y^{-1} X\mu + \text{const} \right) = X^\top \Sigma_y^{-1} X\mu - X^\top \Sigma_y^{-1} y.$$

This is zero iff

$$\mu = \left(X^\top \Sigma_y^{-1} X \right)^{-1} X^\top \Sigma_y^{-1} y.$$

The displayed inverse requires X to have full column rank.

The exponential model and Gamma prior

Observed lifetimes $x_1, \ldots, x_n \geq 0$, with $n \geq 1$. Given λ, the observations are independent and identically distributed.

$$X_i \mid \lambda \sim \exp(\lambda), p(x_i \mid \lambda) = \lambda e^{-\lambda x_i}, \lambda > 0$$

$$\lambda \sim Gamma(\alpha, \beta), \alpha > 0, \beta > 0$$

The hyperparameters are known and fixed: α is the shape and β is the rate.

$$p(\lambda) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha - 1} e^{-\beta\lambda}, \lambda > 0$$

Γ denotes the Gamma function. Both densities are zero outside their support.

Exercise 3(a)

The likelihood of the observed lifetimes

Under the exponential model and Gamma prior just introduced, what is the likelihood of the full dataset?

$$p(x_{1:n} \mid \lambda) = ?$$

Working

$$p(x_{1:n} \mid \lambda) = \prod_{i=1}^{n} p(x_i \mid \lambda)$$

$$= \prod_{i=1}^{n} \lambda e^{-\lambda x_i}$$

$$= \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}$$

$$S_n = \sum_{i=1}^{n} x_i$$

Exercise 3(b)

The posterior distribution

$p(\lambda | x_{1:n}) \sim \text{Gamma}(\alpha + n, \beta + S_n)$

Use the likelihood from part (a) and the Gamma(α, β) prior.

What is the posterior distribution $p(\lambda \mid x_{1:n})$?

Working

$$p(\lambda | x_{1:n}) \propto p(\lambda)\, p(x_{1:n} | \lambda)$$

$$\propto \frac{\beta^\alpha}{\Gamma(\alpha)} e^{-\beta\lambda} \lambda^{\alpha-1} \lambda^n e^{-\lambda S_n}$$

$$\propto e^{-\beta\lambda} \lambda^{\alpha-1} \lambda^n e^{-\lambda S_n}$$

$$\propto \boxed{\alpha}\ e^{-\lambda(\beta + S_n)} \lambda^{\alpha + n - 1}$$

Exercise 3(c)

The MAP estimate of the failure rate

For n ≥ 1, use the posterior from part (b).

What is the maximum a posteriori estimate of λ?

Working

$$\lambda_{MAP} = \underset{\lambda}{\operatorname{argmax}}\ \text{Gamma}(\lambda; \alpha+n, \beta+Sn)$$

$$= \underset{\lambda}{\operatorname{argmax}}\ \underbrace{-\lambda(\beta+Sn) + \log(\lambda)(\alpha+n-1)}_{f(\lambda)}$$

$$\frac{\partial f}{\partial \lambda}(\lambda) = -(\beta+Sn) + \frac{\alpha+n-1}{\lambda}$$

$$\frac{\partial^2 f}{\partial \lambda^2}(\lambda) = -\frac{(\alpha+n-1)}{\lambda^2} < 0$$

$$\lambda_{MAP} = \frac{\alpha+n-1}{\beta+Sn}$$

Exercise 3(d)

Does the prior matter for large samples?

Statement: For sufficiently large n, the MAP and maximum likelihood estimates are approximately equal.

$$\hat{\lambda}_{MAP,n} \approx \hat{\lambda}_{MLE,n}$$

$$\lambda_{MLE} = \arg\max_{\lambda} \lambda^n e^{-\lambda S_n}$$

$$= \arg\max_{\lambda} n \log(\lambda) - \lambda S_n .$$

Working

$$g(\lambda)$$

$$\frac{\partial g}{\partial \lambda}(\lambda) = \frac{n}{\lambda} - S_n$$

$$\frac{\partial^2 g}{\partial \lambda^2}(\lambda) = -\frac{n}{\lambda^2} < 0$$

$$\lambda_{MLE} = \frac{1}{\frac{S_n}{n}} = \frac{1}{\bar{x}_n}$$

Exercise 3(d)

Comparing MAP and maximum likelihood

Explain why the MAP and MLE become close as n grows, with fixed prior parameters.

Working

$$\hat{\lambda}_{MAP} = \frac{\alpha + n - 1}{\beta + S_n} = \frac{\frac{\alpha}{n} + 1 - \frac{1}{n}}{\frac{\beta}{n} + \frac{S_n}{n}}$$

$$\text{n large} \approx \frac{1}{\bar{X}_n} = \hat{\lambda}_{MLE}$$

Online Bayesian linear regression

$$w \sim \mathcal{N}(0, I_d), y_i = x_i^\top w + \varepsilon_i$$

$$\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2), \sigma_n^2 > 0$$

The inputs are fixed and σ_n^2 is known. The noises are mutually independent and independent of w.

$$\mathcal{D}_t = \{(x_i, y_i)\}_{i=1}^t, \mathcal{D}_0 = \varnothing$$

The posterior distribution of the weights

After t observations, $X = X^{(t)}$ and $y = y^{(t)}$.

$$p_t(w) = p(w \mid \mathcal{D}_t) = \mathcal{N}(w; \bar{\mu}_t, \bar{\Sigma}_t)$$

$$\bar{\mu}_t = \left(X^\top X + \sigma_n^2 I_d\right)^{-1} X^\top y$$

$$\bar{\Sigma}_t = \left(\sigma_n^{-2} X^\top X + I_d\right)^{-1} = \sigma_n^2 \left(X^\top X + \sigma_n^2 I_d\right)^{-1}$$

$$\bar{\mu}_0 = 0, \bar{\Sigma}_0 = I_d$$

- D_t in memory $O(td)$
- Inverse computation $O(d^3)$ $O(d^2 t)$

Exercise 4(a)

Data matrices as sums of observations

The ith row of $X \in \mathbb{R}^{txd}$ is x_i^T and $y = (y_1, \ldots, y_t)^T$.

Prove the identity below and derive the analogous expression for $X^T y$.

$$X^T X = \sum_{i=1}^{t} x_i x_i^T$$

Working

x_i rows of

$$\begin{bmatrix} x_1^T \\ \vdots \\ x_t^T \end{bmatrix}$$

$$X^T X = [x_1, \ldots, x_t] = \sum_{i=1}^{t} x_i x_i^T$$

$$X^T y = \sum_{i=1}^{t} y_i x_i$$

Exercise 4(b)

A posterior update with fixed memory

Design an algorithm whose memory does not grow with t, for fixed d.

Give its stored quantities, initialization and update rules. Explain how to recover the posterior mean and covariance.

Working

$$S_{t+a} = \left(X^{(t+a)}\right)^T X^{(t+a)} = \left(X^{(t)}\right)^T X^{(t)} + x_{t+a} x_{t+a}^T$$

$$b_{t+a} = \left(X^{(t+a)}\right)^T y^{(t+a)} = \left(X^{(t)}\right)^T y^{(t)} + x_{t+a} y_{t+a}.$$

Exercise 4(b)

Recovering the posterior parameters

Express the posterior mean and covariance using the quantities stored by the algorithm.

Working

$S_0 = 0 \in \mathbb{R}^{d \times d}$ $\quad b_0 = 0 \in \mathbb{R}^d$ $\quad O(td)$

$S_{t+n} = S_t + x_{t+n} x_{t+n}^T$ $\quad O(d^2)$

$\mu_{t+n} = (S_{t+n} + \sigma_n^2 I_d)^{-1} b_{t+n}$

$\Sigma_{t+n} = (I_d + \sigma_n^{-2} S_{t+n})^{-1}$

$O(d^3)$ in computation

Exercise 4(c)

The Sherman–Morrison identity

Let A be symmetric and invertible, with $1 + x^{\mathsf{T}}A^{-1}x \neq 0$.

$$\left(A + x\,x^{\mathsf{T}}\right)^{-1} = \left(A^{-1} - \frac{\left(A^{-1}x\right)\left(A^{-1}x\right)^{\mathsf{T}}}{1 + x^{\mathsf{T}}A^{-1}x}\right).$$

Prove the identity by multiplying the proposed inverse by $A + xx^{\mathsf{T}}$.

Working

Direct multiplication and scalar cancellation

Since A is symmetric, $(A^{-1}x)^T = x^T A^{-1}$. Also $x^T A^{-1} x$ is a scalar.

$$\left(A + xx^T\right)\left(A^{-1} - \frac{A^{-1}xx^TA^{-1}}{1+x^TA^{-1}x}\right)$$

$$= I_d + xx^TA^{-1} - \frac{1\left(xx^TA^{-1}\right)}{1+x^TA^{-1}x} - \frac{\left(x^TA^{-1}x\right)xx^TA^{-1}}{1+x^TA^{-1}x}$$

$$= I_d + xx^TA^{-1} - \frac{\left(1+x^TA^{-1}x\right)\left(xx^TA^{-1}\right)}{1+x^TA^{-1}x}$$

$$= I_d.$$

$\left(S_t + \sigma_m^2 I_d\right)^{-1}$.

Positive definite

Exercise 4(d)

$\forall x \in \mathbb{R}^d, x \neq 0 \quad 1 + x^T A_t^{-1} x \neq 0$

An update that avoids a new inverse

Modify part (b) so that each observation takes $O(d^2)$ time.

Specify the stored quantities, initialization and update rules. Justify both time and memory costs.

$S_{t+1} + \sigma_n^2 I_d = \underbrace{(S_t + \sigma_n^2 I)}_{A_t} + x_{t+1} x_{t+1}^T$

Working

$M_{t+1} = (A_t + x_{t+1} x_{t+1}^T)^{-1} = M_t - \left(\frac{M_t x_{t+1} (M_t x_{t+1})^T}{1 + x_{t+1}^T M_t x_{t+1}} \right)$

$\mu_{t+1} = M_{t+1} b_{t+1}$

$\Sigma_{t+1} = \sigma_n^2 M_{t+1}$

$\mu_0 = 0 \qquad \Sigma_0 = \sigma_n^2 I$

$O(d^3)$

$O(d^2)$

Exercise 4(d)

Initialization and computational cost

Give the initialization and posterior parameters. Justify the time and memory costs.

Working

The posterior predictive distribution

Using the predictive distribution formula from the lecture:

$$y_\star | x_\star, \mathcal{D}_{t+1} \sim \mathcal{N}\left(m_\star, s_\star^2\right)$$

$$\mathcal{N}\left(x_\star^\top \bar{\mu}_{t+1},\ \sigma_n^2 + x_\star^\top \bar{\Sigma}_{t+1} x_\star\right)$$

$$m_\star = x_\star^\top \bar{\mu}_{t+1}$$

$$s_\star^2 = \sigma_n^2 + x_\star^\top \bar{\Sigma}_{t+1} x_\star$$

A conditional density that stays the same

Let W and Z be continuous random variables. Suppose, for every z,

$$p(W \mid Z = z) = q(W).$$

Show that p(W) = q(W).

Working

Recognizing a one-dimensional Gaussian

Let f be a density on $\mathbb{R}$, with a > 0 and b $\in \mathbb{R}$.

$$f(x) \propto \exp\left(-\frac{1}{2}ax^2 + bx\right).$$

Which distribution is f?

Working

Recognizing a multivariate Gaussian

Let Λ be symmetric positive definite and $b \in \mathbb{R}^d$. Which distribution is f?

$$f(x) \propto \exp\left(-\frac{1}{2}x^\top \Lambda x + b^\top x\right).$$

Deduce the value of the integral, for $m \in \mathbb{R}^d$:

$$\int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}(x-m)^\top \Lambda (x-m)\right) dx\,.$$

Working

The precision matrix determines the covariance

$$m = \Lambda^{-1} b$$

$$-\frac{1}{2} x^\top \Lambda x + b^\top x = -\frac{1}{2}(x-m)^\top \Lambda (x-m) + \frac{1}{2} b^\top \Lambda^{-1} b$$

$$f = \mathcal{N}\left(\Lambda^{-1} b, \Lambda^{-1}\right)$$

$$\int_{\mathbb{R}^d} e^{-\frac{1}{2}(x-m)^\top \Lambda (x-m)} dx = (2\pi)^{d/2} det(\Lambda)^{-1/2}$$

Exercise 1: answer key

Statement	Answer	Reason
1(a): Gaussian marginal	False	Every Gaussian marginal is Gaussian.
1(a): Linear combination	True	A deterministic linear map preserves Gaussianity.
1(b): Normalized expression	False	T given Z has the same N(0, 1) law for every Z.
1(b): Random angle	True	The conditional variance is $\cos^2\alpha + \sin^2\alpha = 1$.
1(b): Sum and difference	False	Their covariance is zero for every allowed α.
1(c): Independent binary	False	Only n Bernoulli parameters are needed.
1(c): Gaussian coordinates	True	Rotate the covariance to a diagonal matrix.